

Heart Disease Prediction System

1. Introduction

Heart disease is one of the leading causes of death worldwide. This project develops a Machine Learning-based Heart Disease Prediction System using Python, Scikit-learn, Flask, HTML, and CSS.

2. Objective

The objective of this project is to build a machine learning model that predicts heart disease based on patient medical parameters and deploy it using a web application.

3. Technologies Used

Python, Pandas, Scikit-learn, Random Forest Classifier, Flask, HTML, CSS.

4. Dataset Description

The dataset contains patient information such as Age, Gender, Blood Pressure, Cholesterol, Heart Rate, Diabetes, Smoking habits, Chest Pain, and the target variable Heart Disease.

5. Methodology

Step 1: Create Dataset
Step 2: Data Preprocessing and Encoding
Step 3: Feature Selection and Target Separation
Step 4: Train-Test Split
Step 5: Data Scaling using StandardScaler
Step 6: Model Training using RandomForestClassifier
Step 7: Model Evaluation
Step 8: Prediction for New Patient

6. Machine Learning Algorithm

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve accuracy and reduce overfitting.

7. Web Application Development

The Flask framework is used to build the web application. Users enter medical details through an HTML form, and the backend predicts heart disease using the trained model.

8. Results

The system provides prediction results such as 'Heart Disease Detected' or 'No Heart Disease' along with accuracy evaluation.

9. Advantages

- Simple and user-friendly interface
- Fast predictions
- Easy to extend with real datasets

10. Limitations

- Small dataset size
- Not trained with real hospital data
- Not suitable for clinical use without validation

11. Future Enhancements

- Use real medical datasets
- Add more health parameters
- Improve UI design
- Deploy on cloud platforms
- Add authentication system

12. Conclusion

This project demonstrates how machine learning can be integrated with a web application to predict heart disease. With real-world data and improvements, it can be enhanced for practical healthcare applications.